



ROHDE & SCHWARZ

Unternehmensbereich
Rundfunk- und Fernsehtechnik

Service Manual

TV Test Receiver

EMFT

821.4019

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1 Opening the Unit

- 1.1 Remove 4 screws (a) at rear of instrument (Fig. 1)
- 1.2 Remove rear feet (b) (Fig. 1)
- 1.3 Remove top part of housing (c), remove bottom part of housing (d) (Fig. 1)
- 1.4 Loosen 2 screws (e), hinge out IF section (f) and latch in place (Fig. 2)
- 1.5 Loosen 2 screws (g), hinge out RF section (h) and latch in place (Fig. 3)
- 1.6 Caution: do not touch between the angle pieces (i) on the righthand side of the instrument with the IF or RF section hinged out.

2 Removal of Power Supply

- 2.1 Pull out power switch push rod (k) (Fig. 3)
- 2.2 Open up plug X21 (l) on motherboard and disconnect (Fig. 3)
- 2.3 Remove screws (m) at rear of instrument (below left foot) (Figs. 1, 4)
- 2.4 Remove screws (n) at righthand side of instrument (Figs. 2, 4)
- 2.5 Place instrument with rear panel protruding beyond the edge of the bench, swing down power supply and remove (Fig. 5).

3 Opening the Front Panels

- 3.1 Remove 2 screws (o), swing out synthesizer front panel to right (Fig. 6)
- 3.2 Remove 2 screws (p), swing out motherboard front panel to left (Fig. 6).

4 Advice on Troubleshooting

For troubleshooting, first use the overall block diagram to find the faulty module (board).

Once the faulty module has been found, reference can be found in the respective summary of circuit documents to the applicable block diagram in Register 3 and possibly to a separate block diagram.

Use the circuit diagram and the detailed description of the module to initially check the input parameters. If these are OK, check the individual functions of the module. If an input signal is missing, look for the source and measure the signal at its output. The block diagrams help to localize a fault.

A block diagram 821.4019 sheet 10.1 in Register 3 is useful for examining the power supply. Every voltage can be followed up to the respective IC with the pin numbers given. Low-value resistors connected in series are not shown and must be taken into account during troubleshooting.

The summary of circuit documents is followed by circuit diagrams, board layouts and components lists.

5 Precautionary Measures with CMOS Circuits

The CMOS circuits used are protected internally against small static discharges by clamping diodes to the positive voltage and to 0 V as well as by series resistors. Contact to ground must always be made before touching the circuits (contact frame or ground on card). Modules or individual components may only be removed with the power supply switched off. Boards must be packed in conductive material during transport.

When carrying out measurements ensure that test prods do not produce short-circuits between neighbouring points of the integrated circuits and other components. Short-circuits could destroy integrated circuits.

It is not admissible to connect external voltages to the outputs of logic circuits. Soldering on PCBs or other components may only be carried out with the instrument switched off. Do not forget the +5 V_{BAT} voltage on components with a battery back-up. Soldering irons with isolating transformers should always be used.

6 ID Numbers

The basic instrument and the individual modules each have their own ID numbers. These numbers are shown on every sheet in the description (mostly in the bottom line) and help to assign individual components to the modules. The instrument model is shown in the 8th and 9th positions of the ID number and is described in the parts list of Register 3.

Basic instrument	821.4019
RF section	821.9010
IF section/video amplifier	821.7518
Synthesizer	821.9710
Display board for synthesizer	821.9862
Motherboard	821.8514
Display board for motherboard	821.8366
Power supply	822.0917
Power supply board	821.4519
IEC/IEEE bus interface (option)	822.1513

EMF ADJUSTMENT INSTRUCTIONS, POWER SUPPLY

Test	Mode	Default setting	Measuring instrument	Signal input	Meas. instr.	Test point	Specific.	Adjustment
Output voltage					14	W21.2	30V $\pm$ 1V	Check
						W21.3	+ 5V > + 3.2V	Check with instrument switched off
						W21.4, 5, 12, 13	+ 12V $\pm$ 0.3V	Adjust using R30
						W21.6,11	+ 5V + 0.2/-0.1V	Adjust using R20
						W21.7,10	Approx. + 5.7V	Check without load
						W21.8,9	-12V $\pm$ 0.3V	Adjust using R40

EMF ADJUSTMENT INSTRUCTIONS, MOTHER BOARD

Test	EMF mode	Default setting	Meas. Instr.	Signal input	Meas. Instr.	Test point	Specific.	Adjustment
Heating of N101					18	Heating block	+ 60°C	Measure heating block temperature, approx. + 60°C
IF / RF crosstalk	IF		8	IF input X5.3 38.9 MHz 200 mV	16	X101		Disconnect cable W101 from X101 and connect analyzer Reference level with IF mode (selectable in RF section or X122:1-2), X113 terminated by 50 Ω (RF section)
	RF						typ. -80dB	Switch to RF mode (RF section or X122:2-3), measure crosstalk X122:1-2, connect cable W101 to X101

EMF ADJUSTMENT INSTRUCTIONS, MOTHERBOARD

Test	EMF mode	Default setting	Meas. Instr.	Signal input	Meas. Instr.	Test point	Specific.	Adjustment
Measurement of input voltage and AFC		R103 → 2/3, R119, R133 → 1/2	1	X5.3 100 mV	1-TK	P101	38.9MHz ± 200 kHz	Adjust bandpass filter using L101, L102
			8	X5.3 38.9 MHz 100 mV	14	P103, P104	0V	Adjust AFC voltage difference between P103 and P104 using L103
				V _{IN} display		100 mV	Adjust display using R133	
				X5.3 38.9 MHz 10 mV		V _{IN} display P102	10 mV	Adjust display using R119 Repeat 10 mV and 100 mV settings until both displays are correct. R103 can be used to set an optimum linearity display if necessary
Typ. values X5.3 / 38.9 MHz: P102 1 mV + 1.2 V 10 mV + 2.1 V 100 mV + 2.9 V 200 mV + 3.2 V 1 V + 3.9 V								

EMF ADJUSTMENT INSTRUCTIONS, MOTHER BOARD

Test	EMF mode	Default setting	Meas. Instr.	Signal input	Meas. Instr.	Test point	Specific.	Adjustment
Display of vision carrier missing			8	X5.3 38.9 MHz 2 mV	14		LED indication	Adjust threshold using R185 such that LED lights up
AF section		R383, R397, R423, R459, R481 → 1/2 R118, R123 (front panel) → remove 1/2 X405, X409, X407, X415						
20-kHz low pass, sound 1			9	X405.2-3 54.688 kHz approx. 300 mV	6	Mono-1 output X8	min.	Adjust to minimum amplitude using L348 Insert X409
60-kHz lowpass, deviation display			9	X405.2-3 54.688 kHz approx. 300 mV	14 or instr.	N271.14	max.	Rotate core of L385 clockwise and then slowly counterclockwise until maximum level is displayed

EMF ADJUSTMENT INSTRUCTIONS, MOTHERBOARD

Test	EMF mode	Default setting	Meas. Instr.	Signal input	Meas. Instr.	Test point	Specific.	Adjustment
20-kHz lowpass, sound 2			9	X407.2-3 54.688 kHz approx. 400 mV	6	Mono-2 output X9	min.	Adjust to minimum amplitude using 371 Insert X415
				approx. 100 mV	6-TK	P149	max.	Adjust to maximum amplitude using L372, L373
AF gain, sound 2			9	X407.2-3 500 Hz 160 mV	10 Instr.	P147 Mono 2-X9	0.5 V + 6 dBm 30 kHz	Adjust using R423 Adjust R123 (front panel) Adjust deviation display using R397 Check loudspeaker
				X407.2 15 kHz 160 mV	10	Mono 2-X9	-7.6 ± 1dBm	Check deemphasis 50 µs Check 30-kHz deviation display Insert X407:1-2

EMF ADJUSTMENT INSTRUCTIONS, MOTHER BOARD

Test	EMF mode	Default setting	Meas. Instr.	Signal input	Meas. Instr.	Test point	Specific.	Adjustment
AF gain, sound 1			9	X405.2-3 500 Hz 160 mV	10 Instr.	P145 Mono 1-X8	0.5 V + 6 dBm 30 kHz	Adjust using R383, Adjust R118 (front panel) Check deviation display Check loudspeaker
			9	X405.2-3 15 kHz 160 mV	10	Mono 1-X8	-7.6 ± 1 dBm	Check deemphasis 50 µs, Check 30-kHz deviation display Insert X405:1-2
Demodulator sound 1		Remove X325, X327	1	X325.2-3 5.5 MHz 20 mV	1-TK	P135 P137	max. max. sym.	L264 → max. at 5.5 MHz L273 → max. at 5.5 MHz L271 → tuning to 5.5 MHz insert X325:1-2, X327:1-2
Demodulator sound 2		Remove X329, X331	1	X329.2 5.74 MHz 20 mV	1-TK	P141 P143	max. max. sym.	L284 → max. at 5.74 MHz L291 → max. at 5.74 MHz L293 → tuning to 5.74 MHz Insert X329:1-2, X331:1-2
Demodulator sound 1		Loosen W103 from X103 Remove X327	MA1	X103 f _{mod} 500 Hz ± 50 kHz deviation f _{mod} 500 Hz ± 30 kHz deviation	6-TK 11 10	P133 P135 Mono 1-X8 P145	150 mV _{pp} 5-6 V _{pp} Type K = 0.1 % 0.5 V _{rms}	L263 → max. at 38.9 MHz L264 → max. level 5.5 MHz L273 → max. display of level, min. distortion L271 → min. distortion, symmetrical S curve R383 Insert X327:1-2 Connect cable W103 to X103

EMF ADJUSTMENT INSTRUCTIONS, MOTHERBOARD

Test	EMF mode	Default setting	Meas. Instr.	Signal input	Meas. Instr.	Test point	Specific.	Adjustment
Demodulator sound 2		Separate W104 from X104 Remove X331	19	X104 f_{mod} 500 Hz $\pm$ 50 kHz deviation	6-TK	P139	max.	L283 → max.at 38.9 MHz, approx.150mV _{pp}
						P141	max.	L284 → max. level 5.74 MHz, 5 - 6 V _{pp}
				11	Mono 2-X9	Type K = 0.1 %	L291 → max. display of level, min. distortion L293 → min distortion, symmetrical S curve	
				10	P147	0.5 V _{rms}	R423	
Identification decoder	Pilot	Remove X331	3	X104 TT 2, stereo ON TT 1 OFF	6-TK	P149	max.	L372, L373 → max. level approx. 70 mV _{pp}
	P151					max.,LED	L374 → max. level approx. 0.15 V _{pp} , LED display	
	Dual sound					P155	"	R481 → max. level at 274.1 Hz approx. 5 V _{pp} , LED display
	Stereo					P153	"	R459 → max. level at 117.5 Hz approx. 5 V _{pp} , LED display Insert X331:1-2 Connect cable W104 to X104

EMF ... ADJUSTMENT INSTRUCTIONS, IF SECTION / VIDEO AMPLIFIER

Test	EMF mode	Default setting	Meas. instr.	Signal input	Meas. instr.	Test point	Specific.	Adjustment
Return loss, 50-Ω IF input	MAN	R102(front panel) → 1/2	1-BR	X5.3 approx. 100mV 30-40 MHz			≥ 20dB	In range 32 to 40 MHz
Return loss, 50-Ω IF output	MAN		1-BR	X5.4 approx. 100mV 30-40 MHz			≥ 12dB	In range 32 to 40 MHz
Return loss, 75-Ω video input		R445(IF sect.), R458 (front panel) → 1/2	2	X6; X4.4 approx. 200mV			≥ 20dB	In range 0 to 6 MHz
Return loss, 75-Ω Q output		R267(mother-board) → 1/2	2	X7; X4.3 approx. 200mV			≥ 20dB	In range 0 to 6 MHz
Video amplifier	ENVEL.	R384, R399, R403, R412, R423, R428 (IF sect.) → 1/2 Insert X365:2-3 Remove X366	2	P364	2	X6		R428 → adjust to flat frequency response (0 dB) in range 0 to 6 MHz Insert X366 R399 → approx. + 0.5 dB boosting at 6 MHz Insert X365:1-2
IF bandpass	ENVEL SOUND TRAP OFF SOUND TRAP ON	Remove X506	1	X506.2-3	1	P361		L551, L552, L553 → adjust to flat frequency response between 33 and 40 MHz C654 → trap at 33.13 MHz (approx. -10 dB) Insert X506:1-2

EMF ... ADJUSTMENT INSTRUCTIONS, IF SECTION / VIDEO AMPLIFIER

Test	EMF mode	Default setting	Meas. instr.	Signal input	Meas. instr.	Test point	Specific.	Adjustment
Nyquist filter	ENVEL SOUND TRAP ON	C618, C616, C622, C623, C634, C631, C636 (IF sect.) → 1/2 Remove X501	1	X505.2-3	1	P361		C623 → 40.2 MHz C631 → 42.75 MHz C616 → approx. 60 MHz Adjust Nyquist slope using C618, C622, C634, C636, 37.4 MHz → 0 dB 38.4 MHz → -2 dB 38.9 MHz → -6 dB 39.4 MHz → -14 dB L171 → adjust trap to 33.3 MHz Insert X505: 1-2
All-pass filter						P171		
	ENVEL SOUND TRAP ON	R559, R563, R562, R564, R577, R591, R594, R593, R595; C583, C581, C604, C602 → 1/2 Remove X504	1	X504.2-3	1	P515		L532 → 35 MHz C604, R594 → adjust to flat frequency response
	SOUND TRAP OFF							C597 → approx. 34.5 MHz C602, R591 → adjust to flat frequency response Insert X504: 1-2
	SOUND TRAP ON	Remove X502		X502.2-3				L522 → 39 MHz C583, R563 → adjust to flat frequency response
	SOUND TRAP OFF							C574 → approx. 37 MHz C583, R559 → adjust to flat frequency response Insert X502: 1-2

EMF ... ADJUSTMENT INSTRUCTIONS, IF SECTION / VIDEO AMPLIFIER

Test	EMF mode	Default setting	Meas. instr.	Signal input	Meas. instr.	Test point	Specific.	Adjustment
Highpass (sound trap)	ENVEL. SOUND TRAP ON	C531, C547, C537, C544, C599 → 1/2 Remove X501: 1-2 and X503	1	X501.2-3	1	P513		C542 → 33.4 MHz C557 → 31.9 MHz L512 → approx. 22 MHz C531, C547, L512 → frequency response in bassband up to 40 MHz C544 → approx. 31.4 MHz C559 → approx. 29.5 MHz C537 → approx. 20 MHz Adjust to flat frequency response between 33.5 and 40 MHz Insert X501:1-2
	SOUND TRAP OFF							
IF control stage (sound trap)	ENVEL. MAN SOUND TRAP ON	R143, R511 (IF sect.) R102 (front panel) → 1/2	1	IF input X5.3	1-TK	P103		Check frequency response then: C508 → trap 33.36 MHz R507 → to max. trap depth C516 → trap 40.4 MHz C512 → approx. 32 MHz
	SOUND TRAP OFF					P513		Check frequency response of sound trap, subsequently insert X503
IF output stage	ENVEL. MAN SOUND TRAP ON		1	IF input X5.3	1	X5.4		33 to 40 MHz ref. 38.9 MHz ± 0.5 dB

EMF ... ADJUSTMENT INSTRUCTIONS, IF SECTION / VIDEO AMPLIFIER

Test	EMF mode	Default setting	Meas. instr.	Signal input	Meas. instr.	Test point	Specific.	Adjustment
Intercarrier processing	ENVEL MAN SOUND TRAP ON	R102 (front panel) R183, R186, R203, R206 → 1/2	1	IF input X5.3	1	P105 P107		L111, L112 → bandpass 38.9 MHz ± 200 kHz R186, C138 → bandpass 33.4 MHz R183 → level 33.4 MHz; + 1 dB at 38.9 MHz R206, C161 → bandpass 33.15 MHz R203 → level 33.15 MHz; + 8 dB at 38.9 MHz
IF total passband characteristic	ENVEL SOUND TRAP ON MAN		1	IF input X5.3 100mV	1	P361		Correct IF passband characteristic IF level at different stages with 38.8-MHz CW signal, adjust V _{VIDEO} display using R186.
Pulse processing	ENVEL SOUND TRAP ON	X131:2-3 (motherboard)	3, 4	IF input X5.3	19 6	P123 (mother-board) P115 P116 P119 P121	15625 Hz	R213 → H oscillator to line frequency X131:1-2; H oscillator should lock <u>Testing of pulse</u> H sampling pulse Sync sampling pulse 15-kHz detection: with sync → H; without sync → L S: neg. sync pulse

EMF ... ADJUSTMENT INSTRUCTIONS, IF SECTION / VIDEO AMPLIFIER

Test	EMF mode	Default setting	Meas. instr.	Signal input	Meas. instr.	Test point	Specific.	Adjustment
Synchronous detector	ENVEL MAN SOUND TRAP OFF	X133: 2-3 R237, R277, R282, R337, R348, R267 (mother-board) → 1/2 S101 position 1 (sampled)	3, 4	IF input X5.3	6	X137.1 P171	300 mV _{pp} min.	<u>Sound trap</u> Adjust using R102 (front panel) Switch off IF vision, modulate TT1 and TT2 with ± 30-kHz deviation, adjust to minimum amplitude using L171 - switch on IF vision
	SYNCHR SOUND TRAP ON		3, 4			P181	0 Hz	<u>Oscillator (cover over oscillator chamber)</u> Modulate SBUF with sawtooth, adjust sawtooth frequency to minimum beat using L201; insert X133: 1-2
Sync demodulator	ZERO REF ON SYNCHR MAN SOUND TRAP ON	S101 pos.1 (sampled) S101 pos. 2 (unsampled)	3	Video signal No 9, IF input X5.3	6	Q output X7		<u>Q signal phase</u> Adjust to steady display using R281 <u>I-Q display (see Fig. 25)</u> (X: I signal 200 mV/cm; Y: Q signal 20 mV/cm) Adjust black level and zero reference to same Y magnitude using R348

EMF ... ADJUSTMENT INSTRUCTIONS, IF SECTION / VIDEO AMPLIFIER

Test	EMF mode	Default setting	Meas. instr.	Signal input	Meas. instr.	Test point	Specific.	Adjustment
Sync demodulator			3, 4	IF input X5.3 100mV		Video output X6	max.	<u>I signal phase</u> Adjust to maximum using R337
						X6 X7	1 V _{pp} 1 V _{pp}	<u>Q signal level</u> Adjust I signal using R102 (front panel) Adjust amplitude of Q signal using R267 (motherboard) X123:1-2
		S101 pos.1 (sampled)				X7		<u>Loop characteristic</u> Adjust to steady display (minimum flickering) using R281
		S101 pos.2 (unsampled)						<u>I-Q display (see Fig. 25)</u> Adjust sync pulse approx. 10 mV below black level using R277, adjust black level and zero reference to same amplitude using R348
		S101 pos.2 (unsampled)				X7		<u>Frequency / phase comparator</u> Adjust R237 such that a Q signal appears without oscillations even with a large attenuation (IF signal $V_{IN} \leq 10$ mV) after switching the instrument off/on

EMF ... ADJUSTMENT INSTRUCTIONS, IF SECTION / VIDEO AMPLIFIER

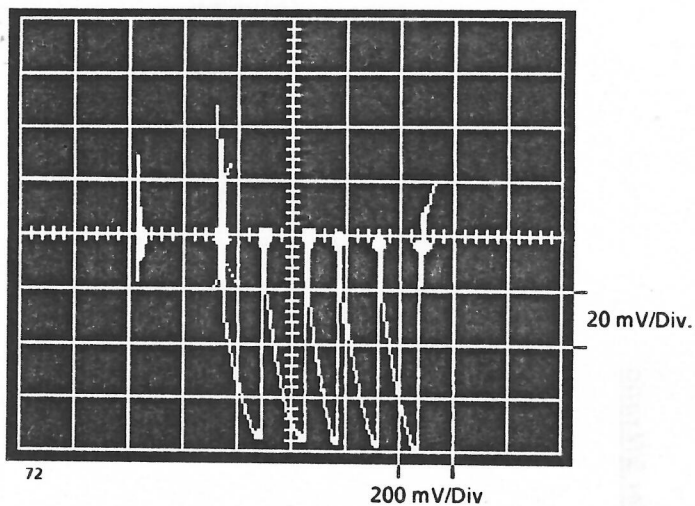
Test	EMF mode	Default setting	Meas. instr.	Signal input	Meas. instr.	Test point	Specific.	Adjustment			
Final adjustment	SYNCHR SOUND TRAP ON		3, 4	IF input X5.3 100mV	6	Video output X6	100 mV	<u>Video level, DC level</u> Adjust CW IF level signal (38.9 MHz) at L110 using R102 (front panel)			
							1 V _{pp}	Modulate SBUF with video, adjust video level using R445.			
							0V	Adjust DC level of zero reference pulse using R423.			
	ENVEL SOUND TRAP ON					-0.1V	Adjust DC level of envelope curve white using R403				
						1 V _{pp}	Adjust level of video signal envelope curve using R412.				
	SYNCHR MAN					V _{VIDEO} display	0	Adjust using R174 (motherboard) without IF input signal			
						V _{VIDEO} display	V _{VIDEO}	Adjust using R186 (motherboard) with IF input signal			
	SYNCHR AUTO					Video output X6, V _{VIDEO} display	1V _{pp}	Adjust R177 such that V _{VIDEO} display corresponds to video signal of 1 V _{pp}			

EMF ... ADJUSTMENT INSTRUCTIONS, IF SECTION / VIDEO AMPLIFIER

Test	EMF mode	Default setting	Meas. instr.	Signal input	Meas. instr.	Test point	Specific.	Adjustment
Final adjustment	SYNCHR; SOUND TRAP, ZERO REF ON						Lines 15 and 328	<u>Zero reference pulse level</u> Adjust zero reference pulse level using R243 (motherboard)
	ENVEL MAN		7	IF input X5.3 100mV	7	P361		<u>Adjustment of IF passband characteristic</u> Attach both covers, sound trap → 33.36 MHz. Coarse adjustment of video frequency response + group delay time. Level approx. 1 dB tolerance, group delay 20 ns
			3	TT 1/2 ± 30kHz deviation, mod.	12	X6		<u>Sound trap</u> Carry out very fine adjustment of video signal and sync pulses for best S/N ratio using C542, C508, C654

EMF ... ADJUSTMENT INSTRUCTIONS, IF SECTION / VIDEO AMPLIFIER

Test	EMF mode	Default setting	Meas. instr.	Signal input	Meas. instr.	Test point	Specific.	Adjustment
Final adjustment	SYNCHR SOUND TRAP ON/OFF		MA3	IF input X5.3	MA3	Video output X6	Data sheet	<u>IF and VF amplitude characteristic</u>
	ENVEL SOUND TRAP OFF							<u>IF and VF group delay</u> Adjust all-pass filter Alternately adjust IF and VF frequency response in synchronous detection
	SYNCHR SOUND TRAP ON AUTO		MA2	IF input X5.3	MA2	X8 sound 1 X9 sound 2		<u>Intercarrier S/N ratio</u> Check



Synchronous - sampled

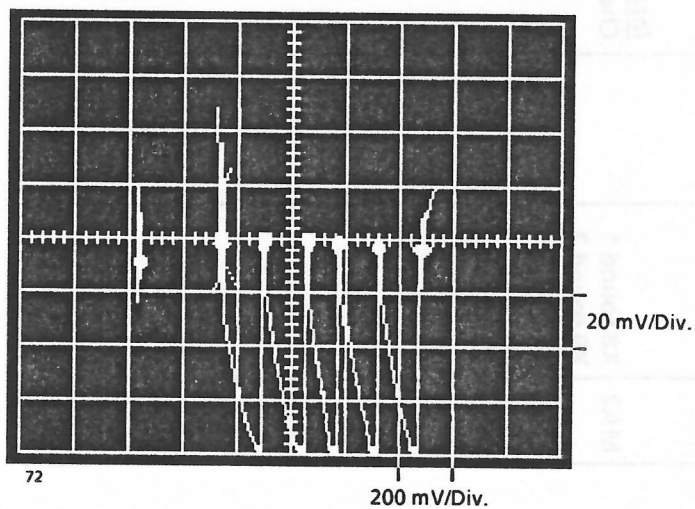
X input → video signal 1 V_{pp}
(staircase)

Y input → Q signal

X-Y display

Zero reference "On"

Fig. 24



Synchronous - non-sampled

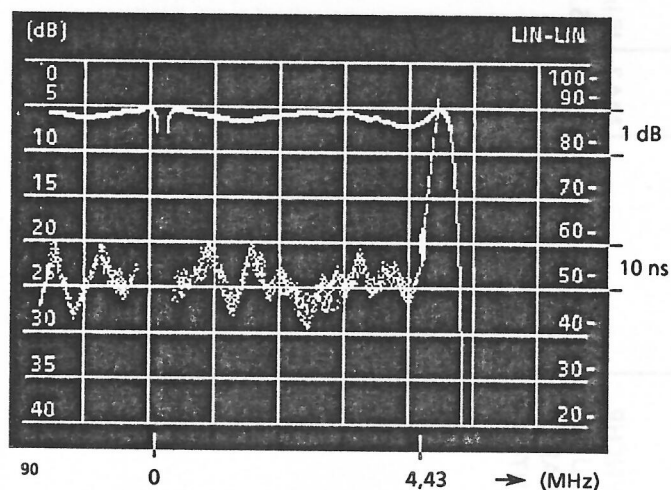
X input → video signal 1 V_{pp}
(staircase)

Y input → Q signal

X-Y display

Zero reference "On"

Fig. 25



Synchronous

IF, VF amplitude
"Sound trap on"

IF, VF group delay
"Sound trap on"

IF input → Q signal

Fig. 26

EMF ADJUSTMENT INSTRUCTIONS, RF SECTION

Test	EMF mode	Default setting	Meas. instr.	Signal input	Meas. instr.	Test point	Specific.	Adjustment
1st oscillator	Channel 2 [air]	X263: 2-3, remove D261 from socket			21	Z310 (0311)		<u>Control voltage</u> Control voltage must be within range 1 to 28 V for all oscillators
		X25:1-2 (synth.) X25:1-2 X25:1-2 X25:2-3						Oscillator 1 (front panel: channel [air] 2-12) Oscill. 2 (front panel: special [CATV std.] 11-41) Oscillator 3 (front panel: channel [air] 21-39) Oscillator 4 (front panel: channel [air] 40-83)
		X25(synth.): 1-2			22	X265	1047 MHz	<u>Oscillator frequency</u> Adjust frequency using C345
					21	X265	≥ 10 dBm	<u>Oscillator level</u> Level for all channels
2nd oscillator		X296: 2-3, remove D295 from socket			22	X293	959.85 MHz	<u>Oscillator frequency</u> Adjust frequency using C422
		X484: 2-3						
		X484: 1-2; X444: 1-2						<u>Frequency range of 1st reference oscillator</u> Repeat both settings several times
		X631: 2-3				X293	959.97 MHz	Adjust using C454
		X631: 2-4				X293	959.73 MHz	Adjust using C458
		X632: open				X293	959.85 MHz	<u>Nominal frequency</u> Adjust frequency using R653

EMFT ADJUSTMENT INSTRUCTIONS, RF SECTION

Test	EMF mode	Default setting	Meas. instr.	Signal input	Meas. instr.	Test point	Specific.	Adjustment
2nd oscillator		X444: 2-3			22	X293	959.913 MHz	<u>Upper frequency limit of search run</u> Adjust frequency using R447 Rotate R641 up to switchover point (measured at P640, -10 V ↔ 10 V)
					21	X293	≥ 7 dBm	<u>Oscillator level</u> Check level
					22	P620	38.9 MHz	<u>Frequency of 2nd reference oscillator</u> Adjust frequency of C601
Preselection	RF 50Ω, MAN, 0 dB Channel 40 [air] Channel 21 [air]	X240: 2-3, X187: 1-2, X261: 1-3, X25 (synth): 1-2	1	X123, level 0.5 V - 25dB (= 0 dB)	1-BR	X262	620 MHz (-0.2 dB)	<u>Highpass ≥ 620 MHz</u> Adjust cutoff frequency by lowering or raising L183 and L185 (soldered); change L183 and L185 by the same amount; see Fig. 1 for passband curve <u>Bandpass filter 470 - 627 MHz</u> Change coupling of L255 to L253 and C256 to L257 by soldering. Initially set both tappings equally far from the ground of L253 and L257; increase the coupling until 3 humps are visible (tighter coupling: tapping points further away from ground of L253 or L257). Continued on next page

EMF ADJUSTMENT INSTRUCTIONS, RF SECTION

Test	EMF mode	Default setting	Meas. instr.	Signal input	Meas. instr.	Test point	Specific.	Adjustment
Preselection	Channel 21 [air]							<u>Bandpass 470 - 627 MHz (continued)</u> Shift the centre hump into the middle of the passband curve by raising or lowering L255; shift the frequency of the passband curve upwards or downwards by raising or lowering L253 and L257 (by same magnitude). Make coupling unsymmetrical by increasing the centre frequencies, i.e. one tapping further away from ground; reduce the ripple to a minimum of 0.5 dB by reducing the coupling
							-0.2 dB	Adjust the passband curve 470 - 630 MHz as in Fig. 2.
	Channel 30 [CATV std.]						-0.2 dB	<u>Bandpass 372 - 477 MHz</u> Similarly to bandpass 470 - 627 MHz, adjust the passband curve 370 - 480 MHz as in Fig. 3 by changing L193, L197, L195 and the two coupling values
	Channel 21 [CATV std.]						-0.2 dB	<u>Bandpass 300 - 379 MHz</u> Similarly to 470 - 627 MHz, adjust the passband curve 300 - 380 MHz as in Fig. 4 by changing L203, L207, L205 and the two coupling values
	Channel 11 [CATV std.]						-0.2 dB	<u>Bandpass 228 - 307 MHz</u> Similarly to bandpass 470 - 627 MHz, adjust the passband curve 230 - 310 MHz as in Fig. 5 by changing L213, L217, L215 and the two coupling values

EMFT ADJUSTMENT INSTRUCTIONS, RF SECTION

Test	EMF mode	Default setting	Meas. instr.	Signal input	Meas. instr.	Test point	Specific.	Adjustment
Preselection	Channel 5 [air]						-0.2 dB	<u>Bandpass 174 - 185 MHz</u> Similarly to bandpass 470 - 627 MHz, adjust the passband curve 174 - 185 MHz as in Fig. 6 by changing L223, L227, L225 and the two coupling values
	Channel 1 [CATV std.]						-0.2 dB	<u>Bandpass 102 - 181 MHz</u> Similarly to bandpass 470 - 627 MHz, adjust the transmission curve 102 - 181 MHz as in Fig. 7 by changing L233, L237, L235 but not the two coupling values
	Channel 2 [air]						-0.2 dB	<u>Lowpass ≤ 109 MHz</u> Adjust the passband curve ≤ 110 MHz as in Fig. 8 using L245 and L247 (screwed in by same amount)
	Channel 2 [air]	X240: 1-2						<u>VHF trap</u> Set trap to 100 MHz using L242, see Fig. 9
	Channel 1 [CATV std.]	X240: 2-3						Set trap to 100 MHz using L242, see Fig. 10
	0 dB 10 dB 20 dB						10 $\pm$ 1dB 0 $\pm$ 1dB -10 $\pm$ 1dB	<u>Passband attenuation</u> Output level for all filters (1 - 7) " "

- 2.5 -

EMFT ADJUSTMENT INSTRUCTIONS, RF SECTION

Test	EMF mode	Default setting	Meas. instr.	Signal input	Meas. instr.	Test point	Specific.	Adjustment
1-GHz bandpass							-0.2 dB 992 - 1000 MHz	If the bandwidth is larger than 8 MHz, bend L270 and L280 further away from L271 and L279 (or vice versa); see above for correction of passband shape Set the passband curve to frequency range 992 - 1000 MHz using C271, C274 and C279. See Fig. 11 for passband curve
1-GHz bandpass	Channel 18 (CATV) RF 50 Ω MAN 0 dB -10 dB OFF;	X187, X261, X263: 1-2; D261 inserted; X264, X271, X279, X294, X296: 1-2; D295 inserted, X297, X299: 1-2; X25 (synth.): 1-2	1	X123, 0.5 V - 25 dB	1-DK	X380		<u>Fine adjustment</u> Repeat coarse adjustment, frequency range (as result of mixing) 279 - 287 MHz; the passband attenuation should remain approximately the same. Screw the additional cover onto the bandpass and correct the offset using C271, C274 and C279; set slight tilt. See Fig. 11 for passband curve
IF bandpass and SAW filter (SAW only standard B/G)	-10 dB OFF; SAW OFF;	X297: 2-4; X299: 2-3 X25 (synth.): 1-2 <u>Only standard B/G:</u> X512, X514, X520: 1-2 R398 $\rightarrow$ 1/2	1	X298, 0.5 V - 25 dB	1-DK	X113	-0.2 dB	<u>Adjustment of IF bandpass:</u> Adjust passband curve to frequency range 32 - 40 MHz using L401 and L402; adjust bandwidth using C403. See Fig. 12 for passband curve

EMF ADJUSTMENT INSTRUCTIONS, RF SECTION

Test	EMF mode	Default setting	Meas. instr.	Signal input	Meas. instr.	Test point	Specific.	Adjustment
IF bandpass and SAW filter (SAW only standard B/G)	-10 dB ON SAW ON							<u>Checking the 10-dB attenuation</u> Passband curve 10 dB less <u>SAW filter</u> Obtain passband curve in line with position "SAW OFF" in the frequency range 35 - 39 MHz using R526, 528. See Fig. 13 for passband curve.
Overall function	Channel 18 (CATV) RF 50 Ω MAN, 0 dB, -10 dB OFF	Jumpers in standard position; X25 (synth.): 1-2; additional cover for 1-GHz bandpass screwed on X25 (synth.): 2-3	1	X123	1-DK 21	 X113	 3 mV 100 mV	<u>Total gain</u> Set sweep tester in manual mode to passband (approx. 285 MHz) Set sweep tester output Adjust level using R398 <u>Readjustment of 1-GHz bandpass</u> Screw on complete cover. Adjust frequency response until it is flat using C271 or C279 (by not more than 1/16 turn). See Fig. 14 for passband curve

EMFT ADJUSTMENT INSTRUCTIONS, RF SECTION

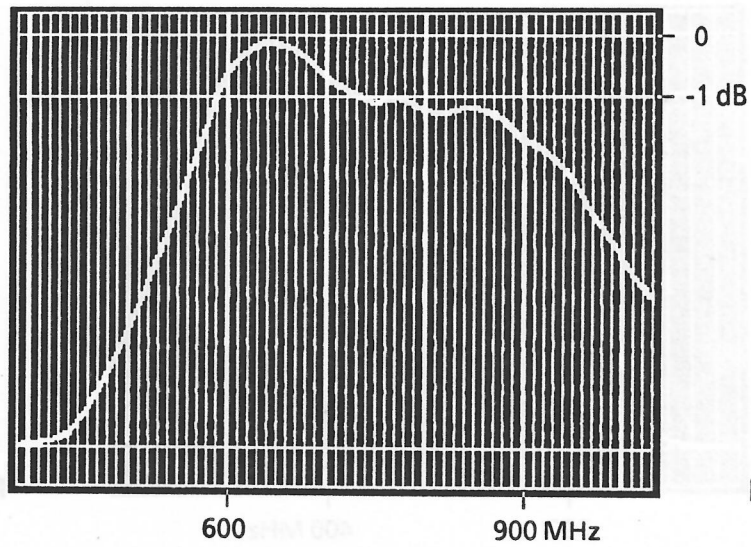


Fig. 1

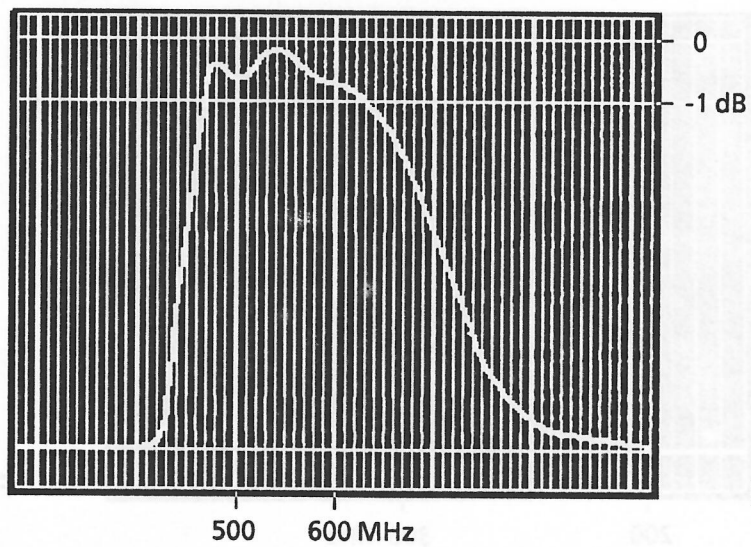


Fig. 2

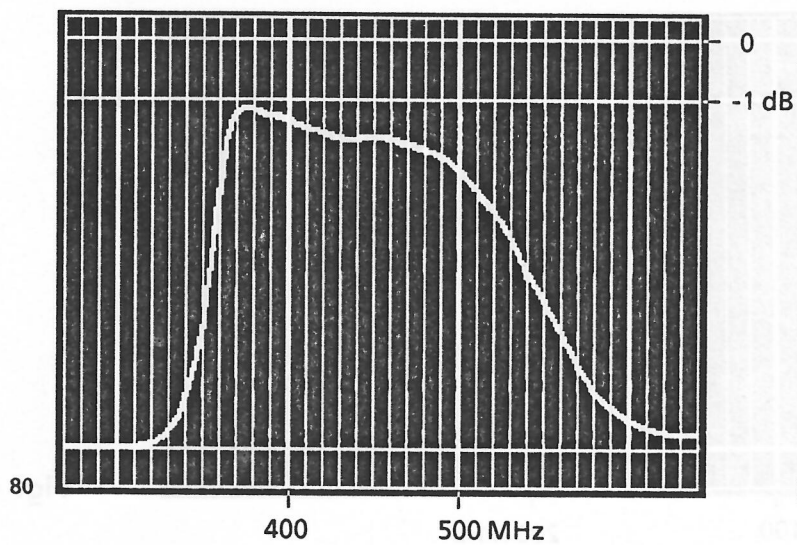


Fig. 3

EMFT ADJUSTMENT INSTRUCTIONS, RF SECTION

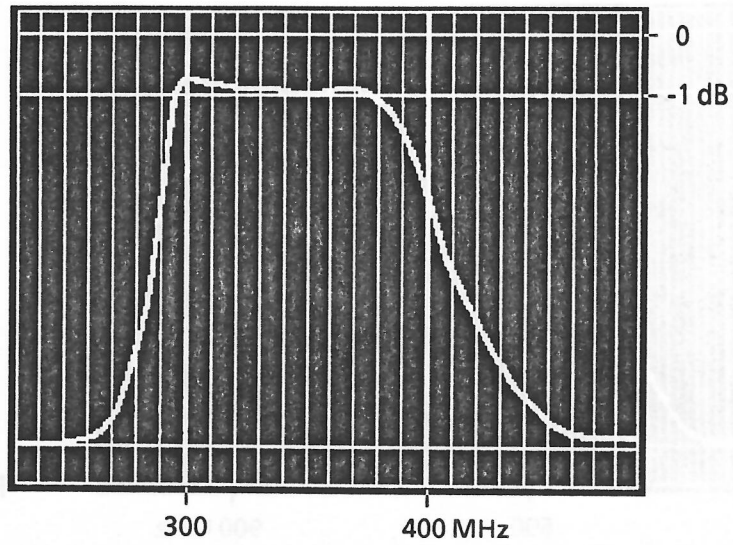


Fig. 4

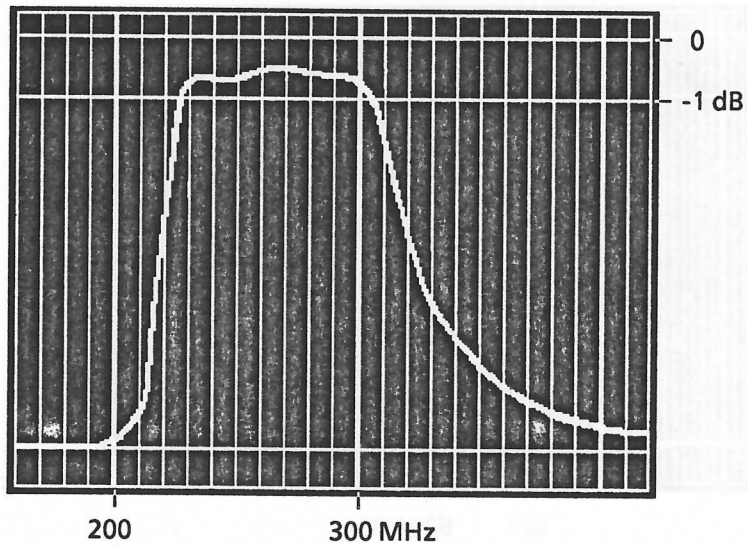


Fig. 5

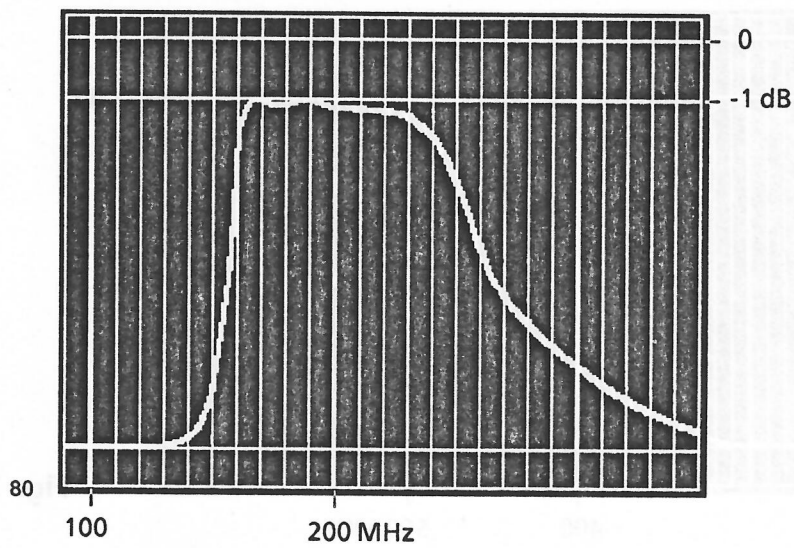


Fig. 6

EMFT ADJUSTMENT INSTRUCTIONS, RF SECTION

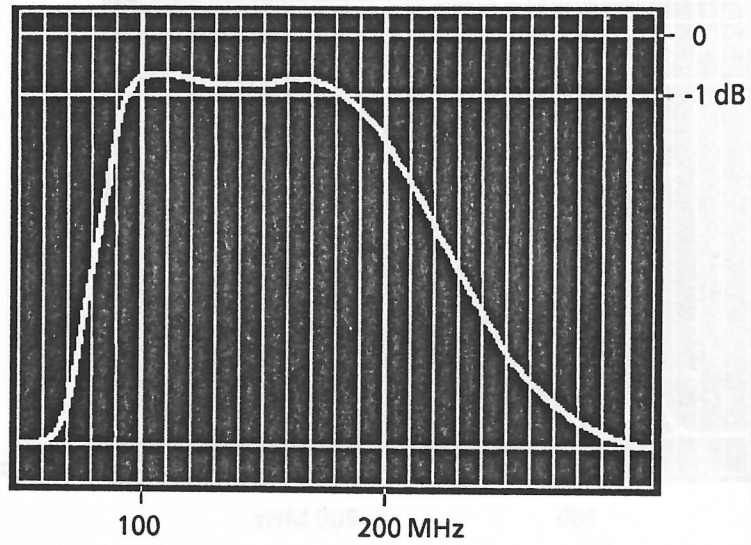


Fig. 7

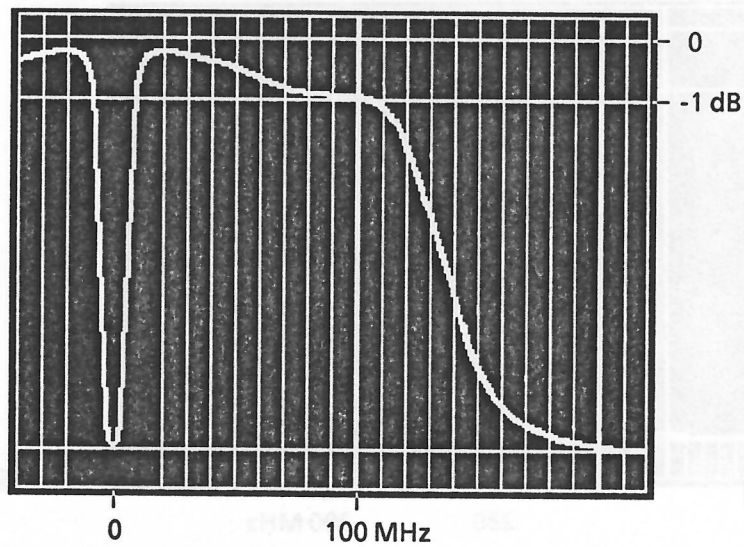


Fig. 8

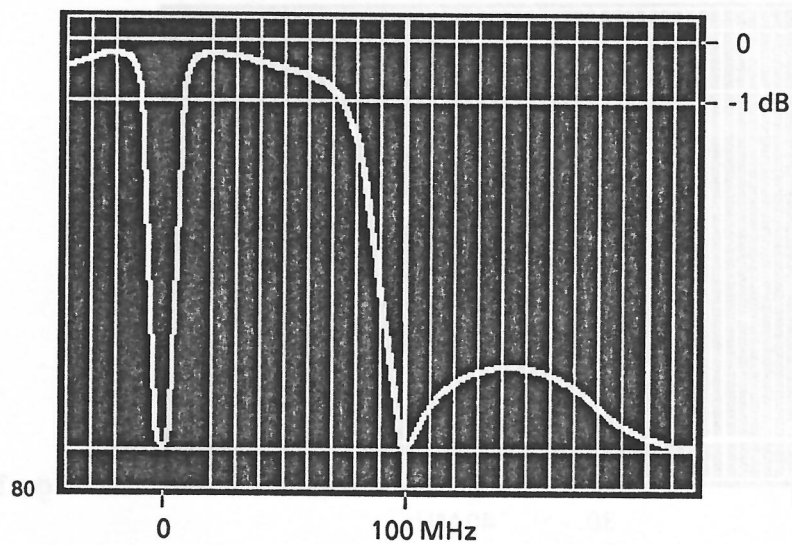


Fig. 9

EMFT ADJUSTMENT INSTRUCTIONS, RF SECTION

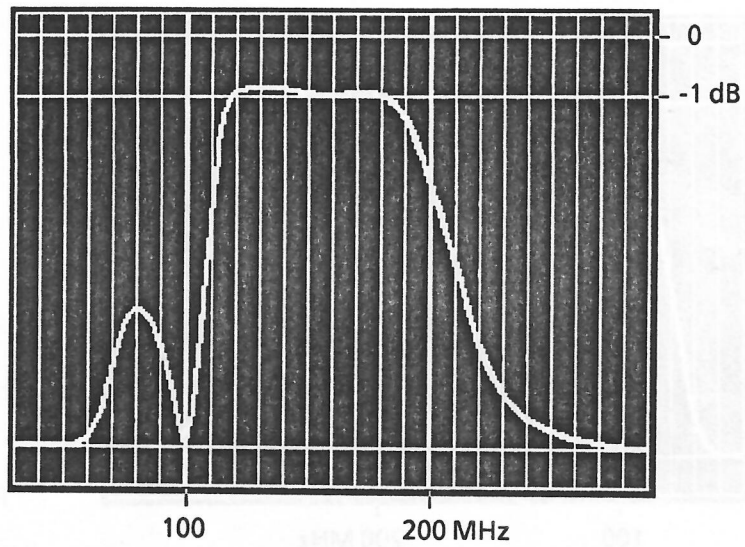


Fig. 10

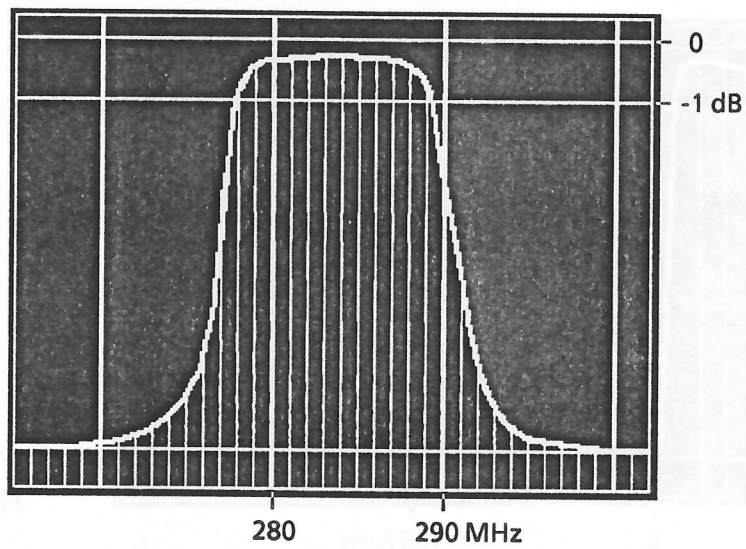


Fig. 11

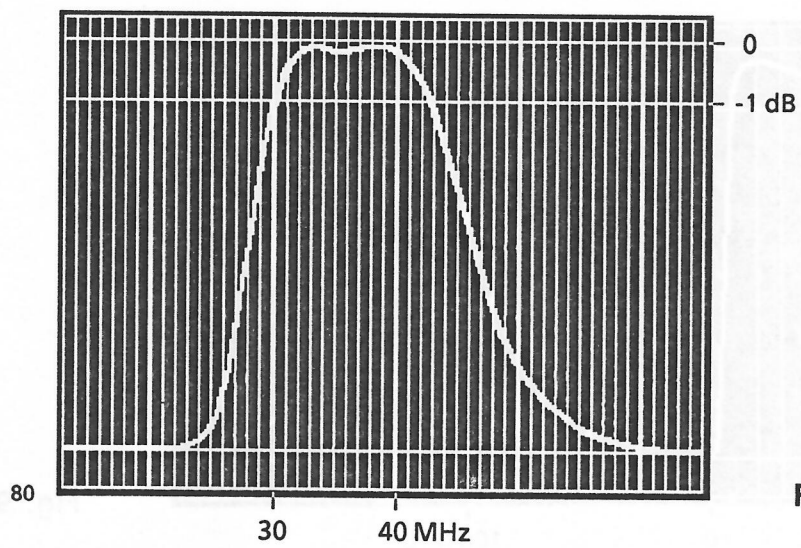


Fig. 12

EMFT ADJUSTMENT INSTRUCTIONS, RF SECTION

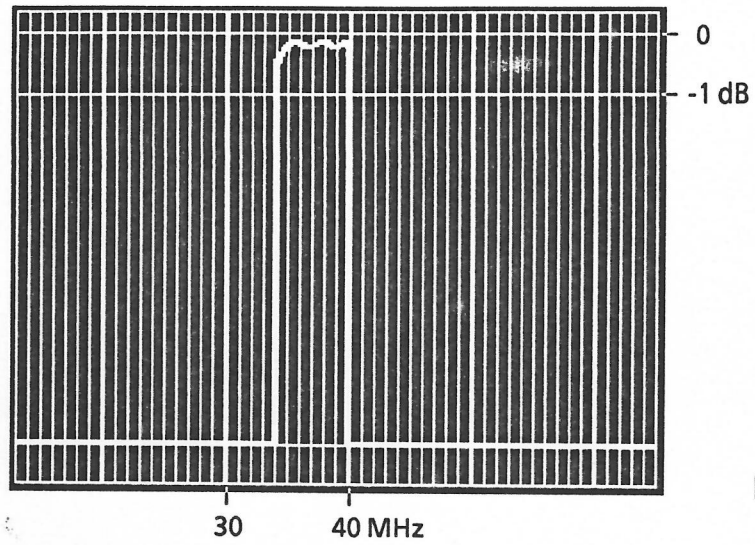


Fig. 13

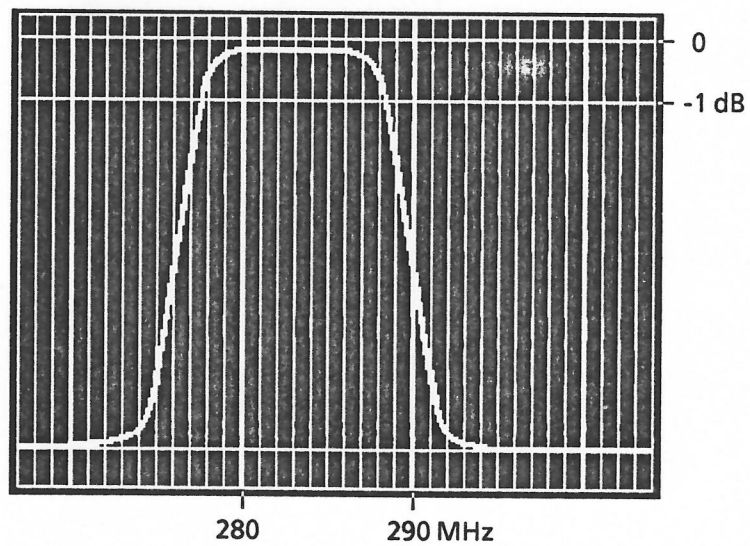


Fig. 14

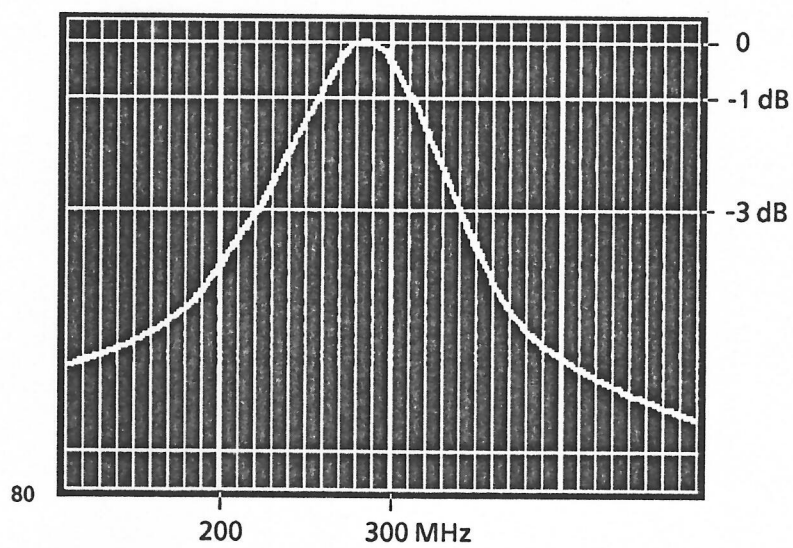


Fig. 15

EMF SYNTHESIZER, ADJUSTMENT INSTRUCTIONS

Test	EMF mode	Default setting	Meas. instr.	Signal input	Meas. instr.	Test point	Specification	Ajustment
Search threshold		S18 → "RF 50Ω" S19 → "0 dB" S21 → "MAN" S22 → "-10dB OFF"	3,4	RF 50Ω-input				<u>Requirement:</u> Complete unit adjusted
		TV channel at approx. 280 MHz		SPF2 signal No.9			approx. 45dBμV	Set search threshold using trimmer on synthesizer board
				Vin 40dBμV Vin 50dBμV				<u>Test:</u> Start search → should not lock in Start search → should lock in on TV channel

